

Two-Sample Problems

1 Two-Sample Location Problem (Shift Model)

One of the most fundamental problems in statistics is comparing two groups:

Do the treatment and control groups come from the same distribution, or is one group systematically shifted upward or downward?

This is known as the *two-sample location problem*.

1.1 Setup and assumptions

We observe two independent random samples:

$$X_1, \dots, X_m \sim F, \quad Y_1, \dots, Y_n \sim G,$$

where

- the X 's might represent outcomes from a **control group**,
- the Y 's might represent outcomes from a **treatment group**.

Typical examples include:

- blood pressure in placebo vs drug group,
- survival times under standard vs new therapy,
- test scores in two different teaching methods.

Key assumptions

- **Independence across and within samples.**

The observations are assumed i.i.d. within each group:

$$X_1, \dots, X_m \stackrel{iid}{\sim} F, \quad Y_1, \dots, Y_n \stackrel{iid}{\sim} G,$$

and the two samples are mutually independent.

This is crucial for the validity of rank-based randomization arguments.

- **Continuity (no ties).**

We often assume F and G are continuous, so ties occur with probability 0. This simplifies the ranking procedure, since each observation has a unique rank.

(With ties, special adjustments are needed; see later remarks.)

- **Location-shift structure.**

The main modeling assumption is that the two groups differ only through a *shift in location*, not through spread or shape.

In other words, the treatment effect moves the distribution left or right but does not change variability.

Why a shift model? In many scientific applications, the primary question is whether the treatment produces a systematic increase (or decrease) in outcomes.

For example:

Does the treatment increase survival times by some constant amount?

Rank-based procedures such as the Wilcoxon test are especially attractive because:

- they do not require normality,
- they are robust to outliers,
- they remain valid for a wide range of continuous distributions.

1.2 Shift model and hypotheses

A location shift model can be written as

$$G(t) = F(t - \Delta),$$

meaning that the treatment distribution is a shifted version of the control distribution.

Equivalently,

$$Y \stackrel{d}{=} X + \Delta.$$

Here Δ is the unknown *shift parameter*:

- $\Delta > 0$ means the treatment tends to produce larger outcomes,
- $\Delta < 0$ means the treatment tends to reduce outcomes,
- $\Delta = 0$ means no shift (no location difference).

For continuous distributions, Δ is often interpreted as the difference in medians (or more generally, a difference in central tendency).

Hypotheses

The standard hypotheses are:

$$H_0 : \Delta = 0 \quad \Longleftrightarrow \quad F(t) = G(t) \quad \forall t,$$

versus

$$H_A : \Delta \neq 0.$$

One-sided alternatives are also common:

$H_A : \Delta > 0$ (treatment tends to increase outcomes),

$H_A : \Delta < 0$ (treatment tends to decrease outcomes).

Remark 1.1 (Model robustness vs interpretation). The Wilcoxon rank-sum test is always a valid test of

$$H_0 : F = G$$

(equal distributions).

However, interpreting it specifically as a *test for a pure shift* is most appropriate when the shift model holds.

If the two groups differ in dispersion or shape, the test may reject even when the medians are equal.

Remark 1.2 (What parameter is being tested?). Under a strict shift model, testing $\Delta = 0$ is equivalent to testing equality of distributions.

Outside the shift model, the Wilcoxon test is often interpreted through the probability-of-superiority parameter:

$$\mathbb{P}(Y > X) = \frac{1}{2},$$

meaning neither group tends to produce systematically larger values. This is the Mann–Whitney interpretation developed below.

2 Wilcoxon Two-Sample Rank-Sum Test (Mann–Whitney)

The most widely used nonparametric procedure for the two-sample location problem is the *Wilcoxon rank-sum test*, also known as the *Mann–Whitney test*.

It replaces raw data values by their ranks, making it robust and distribution-free.

2.1 Ranks and rank-sum statistic

Let $N = m + n$ be the total sample size. Pool all observations into one combined sample:

$$Z_1, \dots, Z_N = (X_1, \dots, X_m, Y_1, \dots, Y_n).$$

Rank them from smallest to largest:

$$1 = \text{smallest}, \quad N = \text{largest}.$$

Let S_j denote the rank of Y_j in the pooled sample.

The Wilcoxon rank-sum statistic is defined as

$$W = \sum_{j=1}^n S_j.$$

Intuition.

- If the treatment group tends to have larger values, then its observations receive larger ranks, so W becomes large.
- If the treatment group tends to have smaller values, then W becomes unusually small.
- Under H_0 , the ranks of X and Y should be thoroughly mixed, so W should fall near its expected value.

2.2 Null distribution and exactness

Assume $H_0 : F = G$ and continuity.

Under H_0 , all N pooled observations are exchangeable, meaning that every way of assigning n ranks to the Y group is equally likely.

There are

$$\binom{N}{n}$$

such assignments.

Therefore, the null distribution of W is:

- **exact**,
- **distribution-free**,
- depends only on (m, n) , not on F .

Exact p-values. For small sample sizes, exact tail probabilities can be computed by:

- enumeration of all rank allocations,
- recursion or dynamic programming algorithms.

This is one of the major advantages of rank-based tests.

2.3 Mean and variance under H_0 (no ties)

Under H_0 and no ties,

$$\mathbb{E}(W) = \frac{n(N+1)}{2}, \quad \text{Var}(W) = \frac{mn(N+1)}{12}.$$

Derivation sketch. Under H_0 , the n ranks assigned to the treatment group form a simple random sample without replacement from $\{1, \dots, N\}$.

Thus W is just the sum of sampled ranks, with known mean and variance.

2.4 Mann–Whitney U statistic and equivalence

An equivalent formulation is the Mann–Whitney statistic:

$$U = \sum_{i=1}^m \sum_{j=1}^n \mathbb{1}(X_i < Y_j).$$

This counts the number of (X, Y) pairs where the treatment observation exceeds the control observation.

Probability of superiority interpretation.

$$\frac{U}{mn} \approx \mathbb{P}(X < Y).$$

Thus the Wilcoxon test can be interpreted as testing whether

$$\mathbb{P}(X < Y) = \frac{1}{2}.$$

This makes the test meaningful even when a pure shift model does not hold.

Equivalence to W . The statistics satisfy the linear relationship:

$$W = U + \frac{n(n+1)}{2}.$$

Therefore, tests based on W and U are exactly equivalent.

2.5 Large-sample approximation

For moderate or large samples, we standardize W :

$$Z = \frac{W - \frac{n(N+1)}{2}}{\sqrt{\frac{mn(N+1)}{12}}} \approx N(0, 1) \quad \text{under } H_0.$$

This yields an approximate p-value using the standard normal distribution.

Continuity correction (optional). Because W is discrete, some implementations subtract or add 0.5 when using the normal approximation.

Ties. With ties, ranks are averaged. The test remains valid, but it is no longer exactly distribution-free, and software uses a tie-adjusted variance for the normal approximation.

2.6 R command

```
wilcox.test(x, y,
            paired = FALSE,
            alternative = "two.sided",
            exact = NULL,
            correct = TRUE)
```

Remark 2.1. In `wilcox.test`, `exact=TRUE` is used when feasible. Ties typically force the use of the large-sample normal approximation.

3 Example: Two-Sample Location Problem (HWC Example 4.1)

Study: transfer of tritiated water across tissue layers in human chorioamnion, comparing term vs gestational age (3–6 months).

Data and test.

```
at.term = c(.80, .83, 1.89, 1.04, 1.45,
            1.38, 1.91, 1.64, .73, 1.46)
gest.age = c(1.15, .88, .90, .74, 1.21)

wilcox.test(at.term, gest.age,
            alternative="greater", conf.int=TRUE)
```

Interpreting typical output.

- $W = 35$ is the sum of ranks of the Y -sample (depending on R's internal convention for which group is called x vs y).
- One-sided p-value ≈ 0.1272 indicates insufficient evidence at 5% that the “term” group tends to have larger values than the gestational-age group.
- The reported CI for the *shift* is a confidence interval for Δ in the shift model.
- The point estimate (about 0.305) is the Hodges–Lehmann estimator (see below).

Remark 3.1 (CI interpretation). The Wilcoxon-based CI is typically a *nonparametric CI for a shift parameter*. Its interpretation is strongest when a shift model is plausible.

4 Hodges–Lehmann Estimator for the Two-Sample Shift

The Wilcoxon rank-sum test provides a powerful distribution-free test for comparing two groups. A natural next question is:

If there is a shift between the groups, how can we estimate its size robustly?

The most widely used robust estimator of a two-sample shift is the *Hodges–Lehmann estimator*.

4.1 Definition via pairwise differences

Recall the shift model:

$$Y \stackrel{d}{=} X + \Delta.$$

A natural way to estimate Δ is to look at all pairwise differences between treatment and control observations:

$$D_{ji} = Y_j - X_i, \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

There are mn such differences.

The Hodges–Lehmann estimator is defined as the median of these differences:

$$\hat{\Delta}_{HL} = \text{median}\{Y_j - X_i : i = 1, \dots, m, j = 1, \dots, n\}.$$

Why is this estimator reasonable? Under the pure shift model,

$$Y_j - X_i \stackrel{d}{=} (X + \Delta) - X = \Delta + (X' - X),$$

so the distribution of $Y - X$ is centered around Δ .

Thus:

- the pairwise differences provide many “noisy” measurements of the shift,
- taking the median extracts the central tendency robustly.

The Hodges–Lehmann estimator is often interpreted as a robust analog of the mean difference:

$$\bar{Y} - \bar{X} \quad (\text{sensitive to outliers}) \quad \longleftrightarrow \quad \hat{\Delta}_{HL} \quad (\text{robust}).$$

Robustness properties.

- Resistant to extreme observations,
- Meaningful for skewed or heavy-tailed distributions,
- Highly efficient under normality (nearly as good as the mean).

Thus $\hat{\Delta}_{HL}$ is one of the most useful nonparametric effect-size estimators.

Link to Wilcoxon inference. The Hodges–Lehmann estimator is closely connected to the Wilcoxon test:

- Wilcoxon tests whether $\Delta = 0$,
- Hodges–Lehmann estimates the magnitude of Δ ,
- Confidence intervals for Δ can be constructed by inversion of the Wilcoxon test.

This produces the standard Wilcoxon/Hodges–Lehmann confidence interval.

4.2 Testing $H_0 : \Delta = \Delta_0$ via a pseudo-sample

To test whether the shift equals some specified value Δ_0 :

$$H_0 : \Delta = \Delta_0,$$

we “remove” the hypothesized shift from the treatment sample:

$$Y'_j = Y_j - \Delta_0, \quad j = 1, \dots, n.$$

Under H_0 , the shifted treatment observations should have the same distribution as X :

$$Y'_j \sim F.$$

Therefore, we can apply the Wilcoxon rank-sum test to

$$X_1, \dots, X_m \quad \text{vs} \quad Y'_1, \dots, Y'_n.$$

By varying Δ_0 , we can determine which shifts are consistent with the data, yielding a confidence interval for Δ .

4.3 Probability of superiority $\delta = \mathbb{P}(X < Y)$

A useful effect-size parameter that does *not* require the shift model is

$$\delta = \mathbb{P}(X < Y).$$

This is called the *probability of superiority* or *common language effect size*.

Interpretation

- Under $F = G$ (continuous),

$$\delta = \frac{1}{2}.$$

- If $\delta > 1/2$, then a randomly chosen treatment outcome tends to exceed a randomly chosen control outcome.
- For example, $\delta = 0.70$ means:

$$P(Y > X) = 0.70,$$

so in 70% of random pairs, treatment beats control.

Thus δ provides an intuitive, probability-based measure of effect size.

Remark 4.1 (Relation to the Mann–Whitney U statistic). The Mann–Whitney statistic

$$U = \sum_{i=1}^m \sum_{j=1}^n \mathbb{1}(X_i < Y_j)$$

counts the number of favorable (X, Y) pairs.

Therefore,

$$\hat{\delta} = \frac{U}{mn}$$

is a natural estimator of δ .

5 Behrens–Fisher Problem (Unequal Variances): Fligner–Policello

The Wilcoxon rank-sum test is most naturally interpreted under a location-shift model. However, in many real applications the two groups may differ not only in location, but also in spread.

This leads to the classical *Behrens–Fisher problem*.

5.1 Problem statement

We observe two independent samples:

$$X_1, \dots, X_m \sim F, \quad Y_1, \dots, Y_n \sim G,$$

with both distributions continuous.

Assume both are symmetric about their medians:

$$\theta_X = \text{median}(X), \quad \theta_Y = \text{median}(Y).$$

The Behrens–Fisher problem is testing

$$H_0 : \theta_X = \theta_Y$$

without assuming equal variances or identical distributional shapes.

Why is this difficult? If the treatment group has larger variance, the Wilcoxon test may reject even if medians are equal, because large values appear more frequently.

Thus we need methods that remain valid when dispersions differ.

Why not Welch’s t ? Welch’s t -test is designed for unequal variances, but it relies on approximate normality.

Fligner–Policello provides a nonparametric alternative that is robust to:

- heavy tails,
- skewness,
- unequal variances.

5.2 Fligner–Policello placements and statistic

Fligner–Policello replaces ranks with *placements*.

Placements

- For each X_i :

$$P_i = \#\{Y_j < X_i\},$$

the number of treatment observations below X_i .

- For each Y_j :

$$Q_j = \#\{X_i < Y_j\},$$

the number of control observations below Y_j .

Thus:

- P_i measures where X_i falls relative to the Y sample,
- Q_j measures where Y_j falls relative to the X sample.

Averages and dispersion

Define mean placements:

$$\bar{P} = \frac{1}{m} \sum_{i=1}^m P_i, \quad \bar{Q} = \frac{1}{n} \sum_{j=1}^n Q_j.$$

Define variability terms:

$$V_1 = \sum_{i=1}^m (P_i - \bar{P})^2, \quad V_2 = \sum_{j=1}^n (Q_j - \bar{Q})^2.$$

Test statistic

The standardized Fligner–Policello statistic is

$$\hat{U} = \frac{\sum_{j=1}^n Q_j - \sum_{i=1}^m P_i}{2\sqrt{V_1 + V_2 + \bar{P}\bar{Q}}}.$$

Interpretation.

- If Y tends to be larger than X , then many X 's fall below Y 's.
- Then $\sum Q_j$ is large and $\sum P_i$ is small.
- This makes $\hat{U}$ positive.

Thus large positive values indicate $\theta_Y > \theta_X$.

5.3 Exact / Monte Carlo / asymptotic computation

Critical values or p-values for $\hat{U}$ can be obtained by:

- **Exact permutation distribution:**

Feasible only when $\binom{m+n}{n}$ is not too large.

- **Monte Carlo permutation:**

Approximate the null distribution by repeatedly shuffling group labels.

- **Asymptotic approximation:**

For large samples,

$$\hat{U} \approx N(0, 1).$$

Fligner–Policello is therefore a powerful extension of rank-based testing to the unequal-variance Behrens–Fisher setting.

6 Example: Behrens–Fisher (HWC Example 4.5: Plasma Glucose in Geese)

This example illustrates the *Behrens–Fisher problem* in a real biomedical setting, where the two groups may differ not only in location but also in variability.

Scientific goal

The goal is to compare plasma glucose concentrations in two groups of Canadian geese:

- **Healthy geese** (control group),
- **Lead-poisoned geese** (exposed group).

Lead poisoning is known to affect metabolism, so researchers want to know whether it is associated with systematically different glucose levels.

Because poisoning may also increase physiological variability, we do *not* want to assume equal variances. Thus this is a natural Behrens–Fisher type setting.

Data

The observed plasma glucose values are:

```
healthy.geese = c(297,340,325,227,277,337,250,290)
lead.poisoned.geese = c(293,291,289,430,510,353,318)
```

Sample sizes:

$$m = 8 \quad (\text{healthy}), \quad n = 7 \quad (\text{lead-poisoned}), \quad N = m + n = 15.$$

A quick look at the treatment group shows some very large values (e.g., 430, 510), suggesting heavier tails or increased spread.

This motivates a robust, nonparametric method.

Fligner–Policello test in R

We apply the Fligner–Policello test using the `NSM3` package:

```
library(NSM3)

pFligPoli(x = healthy.geese,
          y = lead.poisoned.geese,
          method = "Exact")
```

This procedure tests the hypothesis:

$$H_0 : \theta_X = \theta_Y,$$

where θ_X and θ_Y are the population medians, *without assuming equal variances or identical distributional shapes*.

How to interpret the computational choice. The function allows different methods for computing the p-value:

- exact enumeration,
- Monte Carlo permutation,
- asymptotic normal approximation.

Here we choose `method = "Exact"`.

Why is this feasible?

$$\binom{m+n}{n} = \binom{15}{7} = 6435.$$

Since

$$6435 < 10000,$$

it is computationally reasonable to enumerate all possible permutations exactly.

Thus the reported p-value is truly exact, not an approximation.

Results and interpretation

Suppose the output gives the standardized statistic:

$$\hat{U} \approx 1.4676.$$

Direction of the effect

A positive value of $\hat{U}$ suggests that the Y group (lead-poisoned geese) tends to have larger values than the X group (healthy geese).

In other words:

$$\theta_Y > \theta_X \quad \text{appears plausible.}$$

However, the evidence is not overwhelming.

Exact p-values

The exact upper-tail probability is approximately:

$$p_{\text{upper}} \approx 0.0808.$$

The exact two-sided p-value is therefore:

$$p_{\text{two-sided}} \approx 2(0.0808) = 0.1616.$$

Conclusion at common significance levels

- At the 5% level, we do **not reject** H_0 :

$$p = 0.1616 > 0.05.$$

- There is not strong enough evidence to conclude that the medians differ.
- Under the one-sided alternative

$$H_A : \theta_X < \theta_Y,$$

the p-value 0.0808 is sometimes described as *suggestive* or *marginal evidence*, but still not statistically significant at 5%.

Teaching takeaway

This example highlights several important points:

- The two groups may differ in spread, making Wilcoxon shift interpretations questionable.
- Fligner–Policello provides a robust nonparametric test of median equality under unequal variances.
- Exact computation is feasible for small sample sizes and gives reliable inference.
- The lead-poisoned group shows a tendency toward higher glucose, but the evidence is not strong enough to declare significance.

Thus this dataset is an excellent illustration of the Behrens–Fisher problem in practice.

7 Two-Sample Dispersion Problems

So far, we have focused on *location differences* between two groups, such as shifts in medians.

In many applications, however, the main difference is not the center, but the *variability*:

Is one group more spread out or more heterogeneous than the other?

Such questions are called *two-sample dispersion problems*.

Dispersion tests are typically applied after accounting for location (often by centering around the median), because changes in center can mimic changes in spread.

7.1 Dispersion test when medians are equal: Ansari–Bradley

The classical rank-based test for dispersion under equal medians is the *Ansari–Bradley test*.

Setup

Assume two independent continuous samples:

$$X_1, \dots, X_m \sim F, \quad Y_1, \dots, Y_n \sim G.$$

Let

$$\theta_X, \theta_Y \quad \text{denote the medians,}$$

and let

$$\eta_X, \eta_Y \quad \text{denote scale (spread) parameters.}$$

The scientific question is whether the two groups have the same dispersion.

Scale ratio formulation

Conceptually, we define the scale ratio

$$\gamma = \frac{\eta_Y}{\eta_X}.$$

Then the hypotheses are:

$$H_0 : \gamma^2 = 1 \quad (\text{equal dispersion})$$

versus

$$H_A : \gamma^2 \neq 1 \quad (\text{different dispersion}).$$

One-sided alternatives are also common:

$$H_A : \gamma^2 > 1 \quad (Y \text{ is more dispersed}), \quad H_A : \gamma^2 < 1.$$

When is Ansari–Bradley appropriate? The Ansari–Bradley test is most appropriate when:

- the two distributions differ only through a scale change,
- the medians are equal (or have been aligned),
- the distributions are symmetric.

If medians differ substantially, the interpretation becomes problematic, because extreme values could arise from a location shift rather than dispersion.

Ansari–Bradley scoring: center-out ranking

The key idea is to assign scores based on distance from the center.

Pool the samples and order them:

$$Z_{(1)} \leq \cdots \leq Z_{(N)}, \quad N = m + n.$$

Assign symmetric scores:

- score 1 to the smallest and largest observation,
- score 2 to the second smallest and second largest,
- continue inward,
- central observations receive the largest scores.

Thus the scoring emphasizes *how close an observation lies to the center*.

Let R_j be the score assigned to Y_j , and define the Ansari–Bradley statistic:

$$C = \sum_{j=1}^n R_j.$$

Intuition

- If the Y distribution is **less dispersed** (more concentrated), then many Y observations lie near the center, so C tends to be large.
- If the Y distribution is **more dispersed**, then more Y observations appear in the extremes, receiving small scores, so C tends to be smaller.

Thus C measures whether one sample is systematically closer to the center.

Exact vs asymptotic testing In the NSM3 package:

```
pAnsBrad(x = X, y = Y, method = "Exact")  
pAnsBrad(x = X, y = Y, method = "Asymptotic")
```

- Exact enumeration is possible for small samples.
- Large samples use a normal approximation.
- With ties, scores may be averaged, but exact distribution-free validity is lost.

Large-sample mean and variance Let $N = m + n$. Under H_0 ,

$$\mathbb{E}_0(C) = \frac{n(N+2)}{4}, \quad \text{Var}_0(C) = \frac{mn(N+2)(N-2)}{48(N-1)}.$$

Standardizing gives

$$Z = \frac{C - \mathbb{E}_0(C)}{\sqrt{\text{Var}_0(C)}} \approx N(0, 1).$$

7.2 Example: Serum iron determination (HWC Example 5.1)

This example compares the variability of two laboratory methods for measuring serum iron:

- Ramsay method,
- Jung–Parekh method.

The question is whether Jung–Parekh produces more variable measurements.

```
serum = list(  
  ramsay = c(111,107,100,99,102,106,109,108,104,99,  
            101,96,97,102,107,113,116,113,110,98),  
  jung.parekh = c(107,108,106,98,105,103,110,105,104,100,  
                 96,108,103,104,114,114,113,108,106,99)  
)  
  
library(NSM3)  
pAnsBrad(serum$ramsay, serum$jung.parekh, method = "Asymptotic")
```

Interpretation

- Ties are present, so NSM3 uses an asymptotic procedure adapted for ties.
- If $C \approx 234.5$ and the upper-tail p-value is ≈ 0.9093 , there is no evidence that Jung–Parekh is more dispersed.
- A very large p-value suggests the data do not support increased dispersion in the suspected direction.

7.3 Dispersion when medians are unequal: Miller (Jackknife) approach

If the two samples have different medians, dispersion tests that rely on common centering become difficult to interpret.

In this case, Miller’s jackknife approach provides a robust alternative.

Key features:

- Does not require equal medians,
- Works even when distributions differ in location,
- Jackknife quantities remain well-defined even with ties.

In NSM3:

- Compute the statistic with `MillerJack(x=X, y=Y)`.
- P-values may be approximated by:

$$1 - \text{pt}(Q) \quad (\text{small samples}), \quad 1 - \Phi(Q) \quad (\text{large samples}).$$

Thus Miller’s method is useful for dispersion comparisons in more realistic settings.

8 General Two-Sample Distribution Test: Kolmogorov–Smirnov

The Wilcoxon test is primarily sensitive to location shifts. Dispersion tests focus only on scale. Sometimes we want a completely general test:

Do the two samples come from the same distribution at all?

This motivates the Kolmogorov–Smirnov (KS) test.

8.1 Setup

We test equality of the full distributions:

$$H_0 : F(t) = G(t) \quad \forall t$$

versus

$$H_A : F(t) \neq G(t) \text{ for at least one } t.$$

KS is sensitive to differences in:

- location,
- scale,
- skewness,
- general shape.

8.2 Empirical distribution functions

Define empirical cdfs:

$$F_m(t) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}(X_i \leq t), \quad G_n(t) = \frac{1}{n} \sum_{j=1}^n \mathbb{1}(Y_j \leq t).$$

These are step functions estimating F and G .

The KS test compares the maximum vertical distance between them.

8.3 KS statistic

Let $Z_{(1)} \leq \dots \leq Z_{(N)}$ be pooled order statistics, $N = m + n$, and let $d = \gcd(m, n)$.

Define

$$J = \frac{mn}{d} \max_{i=1, \dots, N} |F_m(Z_{(i)}) - G_n(Z_{(i)})|.$$

Large values of J indicate the two empirical cdfs differ substantially.

Reject H_0 if

$$J \geq j_\alpha,$$

where j_α is the critical value under the null.

Because the statistic is discrete for fixed (m, n) , not every significance level α is exactly attainable.

8.4 Exact and asymptotic implementations

```
pKolSmirn(x = X, y = Y, method = "Exact")  
pKolSmirn(x = X, y = Y, method = "Asymptotic")
```

Remark 8.1 (Ties). The KS statistic is still definable with ties, since empirical cdfs remain valid. However, exact null distributions rely on continuity assumptions. Many software packages issue warnings or use approximate corrections when ties occur.

8.5 Large-sample approximation

Define the rescaled statistic

$$J^* = \frac{d}{\sqrt{mnN}} J.$$

As $\min(m, n) \rightarrow \infty$, J^* converges to the Kolmogorov distribution.
Thus we reject if

$$J^* \geq q_\alpha,$$

where q_α is the Kolmogorov critical value.

9 Example: KS Test (HWC Example 5.4: Feedback and Salivation Rate)

This example illustrates the Kolmogorov–Smirnov (KS) test as a *general two-sample distributional comparison*.

Scientific goal

Researchers are interested in whether providing **feedback** affects salivation rate.

Two independent groups are observed:

- subjects receiving feedback,
- subjects receiving no feedback.

Unlike the Wilcoxon test (focused mainly on location shifts), the KS test asks a broader question:

Do the two groups come from the same distribution at all?

Thus KS can detect differences in:

- median/location,
- dispersion/scale,
- skewness,
- more complex shape differences.

Data

The observed changes in salivation rate are:

```
Table5.7 = list(  
  feedback = c(-0.15, 8.6, 5, 3.71, 4.29,  
               7.74, 2.48, 3.25, -1.15, 8.38),  
  no.feedback = c(2.55, 12.07, 0.46, 0.35, 2.69,  
                  -0.94, 1.73, 0.73, -0.35, -0.37)  
)
```

Sample sizes are equal:

$$m = n = 10, \quad N = 20.$$

Because the sample size is small, it is natural to compute an *exact* KS p-value.

Exact KS test in R

Because the sample sizes are small ($m = n = 10$), it is preferable to use the *exact* Kolmogorov–Smirnov test rather than relying on a large-sample approximation.

We can compute the exact KS p-value in R using the NSM3 package:

```
library(NSM3)  
  
pKolSmirn(x = Table5.7$feedback,  
          y = Table5.7$no.feedback,  
          method = "Exact")
```

What is being computed? The KS test compares the two empirical distribution functions (ECDFs)

$$F_m(t) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}(X_i \leq t), \quad G_n(t) = \frac{1}{n} \sum_{j=1}^n \mathbb{1}(Y_j \leq t).$$

The KS statistic is defined as the largest vertical distance between these two step functions:

$$J = \frac{mn}{d} \max_t |F_m(t) - G_n(t)|, \quad d = \gcd(m, n).$$

Here, Multiplying by $\frac{mn}{d}$ takes the raw ECDF gap and expresses it on a common integer grid, reflecting the discrete nature of the data and enabling exact reference distributions.

Interpretation.

- If the two samples come from the same distribution, then the ECDFs should be close, so J will be small.
- If the distributions differ (in location, spread, or shape), then the ECDFs will separate, and J will be large.
- The **Exact** option computes the p-value from the exact null distribution of J , which is especially appropriate when m and n are small.

Thus the KS test provides a fully nonparametric way to detect any difference between the two underlying distributions.

Results and interpretation

Suppose the output gives:

$$J = 6, \quad p_{\text{exact}} \approx 0.0524.$$

What does this mean?

- The p-value is slightly above 0.05, so we do not reject at the strict 5% level.
- However, the result is very close to significance, so the evidence is often described as *borderline* or *suggestive*.
- The data provide some indication that feedback may change the salivation distribution, but the sample size is too small for a decisive conclusion.

Important next step: visualization. Because KS is based on the maximum distance between ECDFs, a very useful diagnostic is to plot the two empirical CDFs:

- Where do the curves separate most?
- Is the difference mainly in the upper tail?
- Is one group more variable?
- Is there a location shift or a shape difference?

Graphical inspection often provides more insight than the p-value alone.

Large-sample approximation

For comparison, one can also compute the asymptotic KS p-value:

```
pKolSmirn(x = Table5.7$feedback,
          y = Table5.7$no.feedback,
          method = "Asymptotic")
```

Suppose this yields:

$$J^* \approx 1.3416, \quad p_{\text{asympt}} \approx 0.0546.$$

Conclusion. The asymptotic approximation leads to essentially the same interpretation:

- evidence is borderline,
- conclusions are similar to the exact test,
- with such small samples, exact computation is preferred when feasible.

Thus the KS test suggests possible distributional differences, but not strong enough evidence at the 5% threshold.

10 Summary and practical workflow

We have now developed several important nonparametric tools for two-sample inference.

Tests covered.

- **Location shift with similar shapes:**

Wilcoxon rank-sum test (Mann–Whitney), best when groups differ mainly in median.

- **Location comparison with unequal dispersion:**

Fligner–Policello test, a nonparametric Behrens–Fisher solution.

- **Dispersion comparison with equal medians:**

Ansari–Bradley test, appropriate when distributions differ only by scale.

- **Dispersion comparison with unequal medians:**

Miller jackknife approach, useful when centering is not comparable.

- **Any distributional difference:**

Kolmogorov–Smirnov test, sensitive to shifts, scale changes, and shape differences.

Recommended diagnostic workflow. In practice, two-sample problems should not be approached with a single test automatically. A sensible workflow is:

- **Step 1: Exploratory plots.**

Always begin with graphics:

- side-by-side boxplots,
- density estimates,
- empirical CDF (ECDF) plots.

- **Step 2: Clarify the scientific question.**

Are we interested in:

- a median shift?
- increased variability?
- any difference at all?

- **Step 3: Choose an appropriate test.**

- If interest is primarily in *location* and spreads look similar: use Wilcoxon rank-sum.
- If spreads differ but medians are still the target: use Fligner–Policello.
- If interest is in *dispersion/precision*: use Ansari–Bradley (equal medians) or Miller (unequal medians).
- If you want a fully general comparison of distributions: use KS, often paired with ECDF plots.

Remark 10.1 (Effect sizes). For two-sample comparisons, p-values only address evidence against H_0 ; they do not quantify the magnitude of the effect.

Common nonparametric effect sizes include:

$$\hat{\Delta}_{HL} \quad (\text{Hodges-Lehmann shift estimate}),$$

$$\hat{\delta} = \frac{U}{mn} \quad (\text{probability of superiority}).$$

Reporting effect sizes alongside p-values provides a much more complete summary.